

B.TECH 1ST YEAR · ENGINEERING CHEMISTRY

Unit III

Polymers & Fuel Chemistry

Subject: Engineering Chemistry

Unit: III — Polymers and Fuel Chemistry

Level: B.Tech — 1st Year

Topics: Polymerization · Thermoplastics · Elastomers · Fuels · Coal · Petroleum

Prepared as per University Syllabus / Exam-Oriented Study Material

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Scope of Unit III: This unit covers the chemistry and applications of polymers (natural and synthetic), polymerization mechanisms, important commercial plastics and elastomers, and the chemistry of fuels including solid, liquid, and gaseous fuels, their calorific values, analysis methods, and alternative energy sources.

1 Introduction to Polymers

1.1 Definition

A **polymer** is a large macromolecule formed by repetitive joining of small structural units called **monomers** through covalent bonds. The word "polymer" is derived from the Greek words *poly* (many) and *meros* (parts).

1.2 Key Terms

- **Monomer:** The small repeating unit that combines to form a polymer. Example: Ethylene ($\text{CH}_2=\text{CH}_2$).
- **Degree of Polymerization (n):** The number of monomer units in a polymer chain. Higher 'n' = higher molecular weight.
- **Repeat Unit (Mer):** The smallest structural unit that repeats throughout the polymer chain.
- **Molecular Weight:** Polymer molecular weight = molecular weight of repeat unit $\times$ degree of polymerization.
- **Polymerization:** The process by which monomers are linked to form a polymer.

Molecular Weight of Polymer = $M_0 \times n$

Where: M_0 = Molecular weight of repeat unit

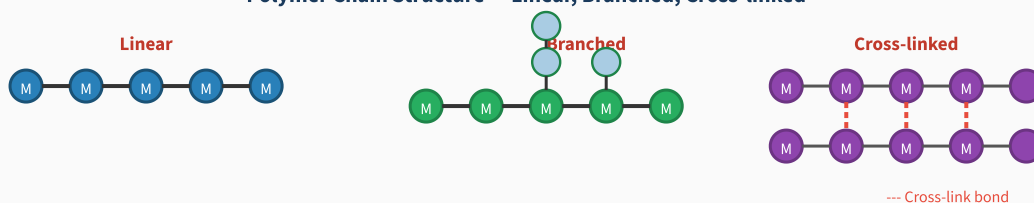
n = Degree of polymerization

1.3 Classification of Polymers

Basis	Type	Examples
Source	Natural / Synthetic	Rubber, Starch / Nylon, PVC
Structure	Linear / Branched / Cross-linked	HDPE / LDPE / Bakelite
Thermal Behaviour	Thermoplastic / Thermosetting	PVC / Epoxy
Polymerization	Addition / Condensation	Polythene / Nylon
Application	Plastics / Elastomers / Fibres	PS / Buna-S / Terylene

1.4 Polymer Chain Structure

Polymer Chain Structure — Linear, Branched, Cross-linked



M = Monomer unit; Blue = Linear; Green = Branched; Purple = Cross-linked

Fig. 1.1 — Types of polymer chain structures

1.5 Properties of Polymers (General)

- **High Molecular Weight:** Range from thousands to millions of g/mol.
- **Mechanical Properties:** Tensile strength, elasticity, and hardness vary with structure.
- **Thermal Properties:** Show glass transition temperature (T_g) and melting temperature (T_m).
- **Chemical Resistance:** Most polymers resist acids and alkalis due to C–C backbone.
- **Electrical Properties:** Most are insulators; some (e.g., polyacetylene) are semiconductors.

★ **EXAM TIP:** Learn the classification table and the difference between monomer, repeat unit and polymer chain — frequently asked in 2-mark questions.

2 Functionality of Monomers

2.1 Definition of Functionality

The **functionality** of a monomer is the number of reactive sites (functional groups or double bonds) available per monomer molecule for bond formation during polymerization.

Functionality (f) determines whether a monomer can form a linear, branched, or network polymer.

2.2 Types Based on Functionality

Functionality	Type	Product	Example
$f = 1$	Monofunctional	No polymer (chain terminates)	CH_3OH (Methanol)
$f = 2$	Bifunctional	Linear polymer	Ethylene, $\text{HO}-(\text{CH}_2)_6-\text{COOH}$
$f = 3$	Trifunctional	Branched / network polymer	Glycerol
$f \geq 3$	Polyfunctional	Cross-linked network (3D)	Phenol ($f=3$)

2.3 Average Functionality

For a mixture of monomers in step-growth polymerization:

$$f_{\text{avg}} = (\text{Sum of } f \times N) / (\text{Total number of molecules})$$

Where: f = functionality of each monomer

N = number of moles of that monomer

Example: 1 mol A ($f=2$) + 1 mol B ($f=2$)

$$f_{\text{avg}} = (2 \times 1 + 2 \times 1) / (1 + 1) = 2.0 \rightarrow \text{Linear polymer}$$

Example: 2 mol A ($f=2$) + 1 mol B ($f=3$)

$$f_{\text{avg}} = (2 \times 2 + 3 \times 1) / (2 + 1) = 7/3 = 2.33 \rightarrow \text{Branched polymer}$$

2.4 Role in Polymerization

- **$f = 2$ (bifunctional):** Only linear chains form. Examples: nylon, polyester.
- **$f > 2$ (polyfunctional):** Cross-linked networks form. Example: phenol-formaldehyde (Bakelite).
- **Critical gelation point:** When f_{avg} exceeds 2, gelation occurs — the system transitions from viscous liquid to an elastic gel (network polymer).

Monomer Functionality and Resulting Structure



Fig. 2.1 — Effect of monomer functionality on polymer structure

★ **EXAM TIP:** Average functionality formula is a common numerical question. Practice the formula with mixed monomer ratios.

3 Polymerization Mechanisms

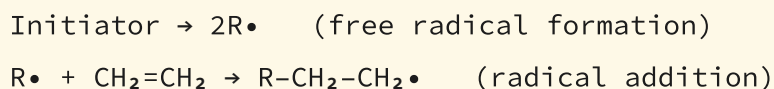
3.1 Chain Growth (Addition) Polymerization

Chain growth polymerization involves the rapid sequential addition of monomer units to an active centre (radical, cation, or anion). Monomers must contain a C=C double bond. No by-product is formed.

MECHANISM — THREE STAGES

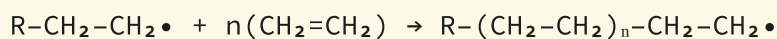
Stage 1 — Initiation

An initiator (e.g., benzoyl peroxide) decomposes to form free radicals. The radical attacks the monomer double bond, creating a monomer radical.



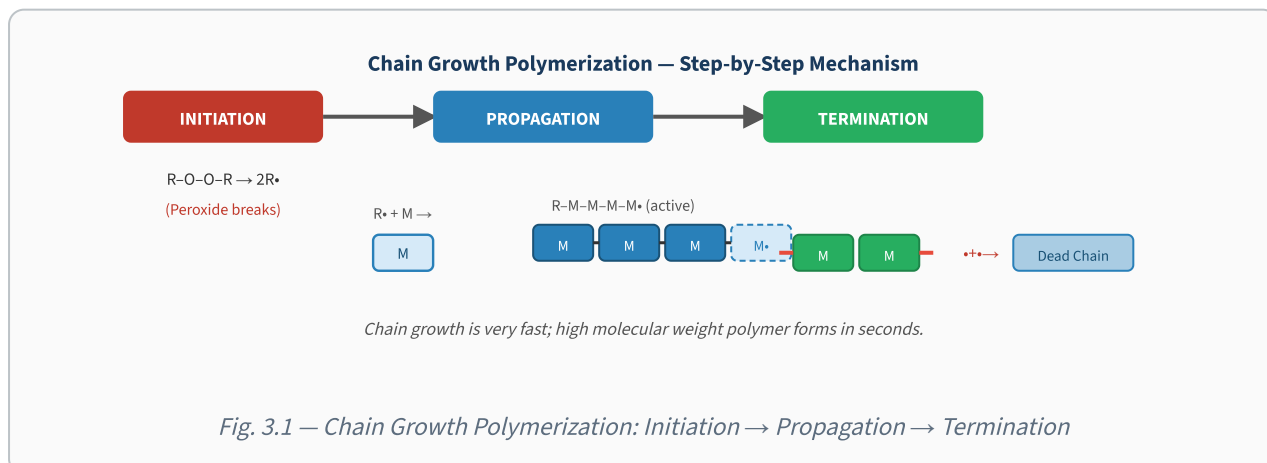
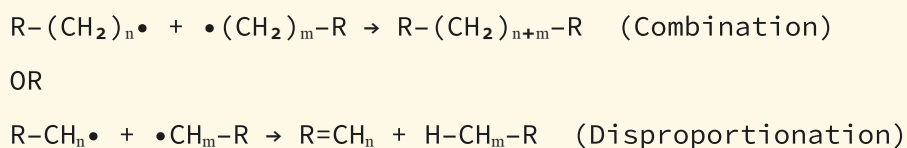
Stage 2 — Propagation

The monomer radical rapidly attacks more monomers, extending the chain.



Stage 3 — Termination

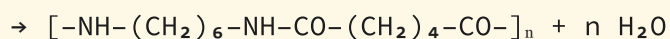
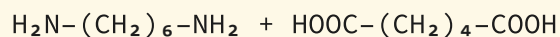
Two active chains combine (combination) or disproportionate to end the chain growth.



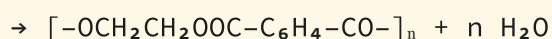
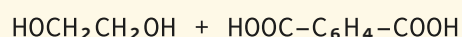
3.2 Step Growth (Condensation) Polymerization

In step-growth polymerization, bifunctional (or polyfunctional) monomers react with each other to form dimers, trimers, and so on, progressively building the polymer chain. A small molecule (e.g., H_2O , HCl) is eliminated as a by-product.

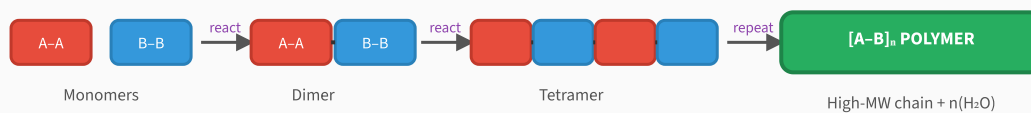
Nylon 6,6:



Polyester (PET):



Step Growth Polymerization — Progressive Chain Building



Step growth is slow; intermediate oligomers exist; high conversion needed for high MW.

Fig. 3.2 — Step Growth Polymerization: Monomers → Dimer → Tetramer → Polymer

3.3 Comparison: Chain Growth vs Step Growth

Property	Chain Growth (Addition)	Step Growth (Condensation)
Monomer type	Unsaturated (C=C)	Bifunctional (–OH, –COOH, –NH ₂)
By-product	None	Small molecule (H ₂ O, HCl)
Rate	Very fast (seconds)	Slow (hours)
MW change	Rapid increase	Gradual increase
Initiator needed	Yes (radical/ionic)	No (catalyst/heat)
Examples	PE, PVC, PS, Buna-S	Nylon, Bakelite, Polyester
Mechanism	Initiation-Propagation-Termination	Stepwise condensation

Thermoplastics and Thermosetting Plastics

4.1 Thermoplastics

Thermoplastics are polymers that soften on heating and harden on cooling, repeatedly. They have linear or slightly branched chains held by weak van der Waals forces. No chemical change occurs during moulding.

- **Structure:** Linear or branched — no cross-links.
- **Behaviour:** Remoldable and recyclable.
- **Solubility:** Soluble in organic solvents.
- **Examples:** PVC, Polystyrene, Polythene, Nylon, Teflon, ABS.
- **Applications:** Packaging, pipes, containers, fibres, films.

4.2 Thermosetting Plastics

Thermosetting plastics undergo an irreversible chemical change (curing/cross-linking) when heated, forming a rigid, infusible, insoluble three-dimensional network. Once set, they cannot be remelted.

- **Structure:** Highly cross-linked 3D network.
- **Behaviour:** Hard, rigid, not recyclable.
- **Solubility:** Insoluble in all solvents.
- **Examples:** Bakelite, Epoxy resin, Urea-formaldehyde, Melamine.
- **Applications:** Electrical fittings, adhesives, laminates, paints.

Thermoplastic vs Thermosetting — Structural Comparison

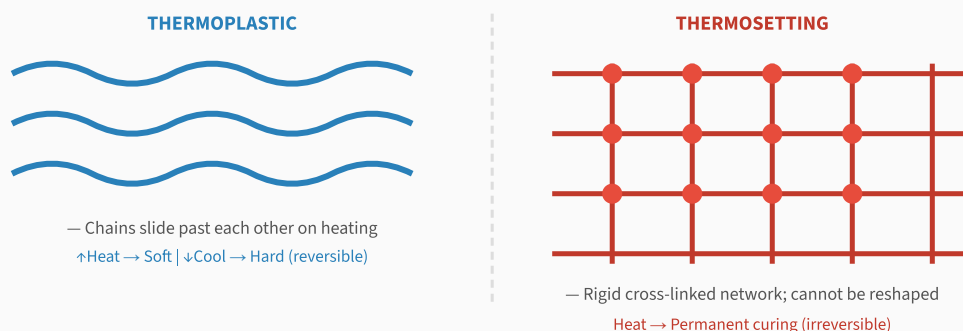


Fig. 4.1 — Structural comparison of Thermoplastic and Thermosetting polymers

4.3 Comparison Table

Property	Thermoplastics	Thermosetting Plastics
Structure	Linear/Branched	Cross-linked (3D network)
On heating	Softens and melts	Chars/decomposes
Recyclability	Recyclable	Not recyclable
Solubility	Soluble in solvents	Insoluble
Strength	Lower	Higher
Moulding	Reversible	Irreversible
Cost	Generally lower	Higher
Examples	PVC, PS, Nylon, LDPE	Bakelite, Epoxy, Urea-F

★ **EXAM TIP:** The difference between thermoplastics and thermosetting is a very common 5-mark question. Focus on structure, behaviour on heating, recyclability, and examples.

5 Polystyrene (PS)

5.1 Introduction

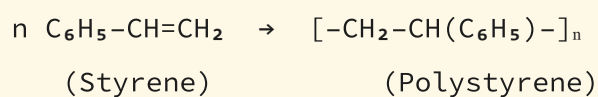
Polystyrene is a thermoplastic polymer produced by chain-growth (addition) polymerization of styrene (vinylbenzene). It is one of the most widely used synthetic plastics.

5.2 Preparation

Monomer: Styrene ($\text{C}_6\text{H}_5\text{--CH=CH}_2$)

Initiator: Benzoyl peroxide or AIBN (radical initiator)

Condition: Heat / UV light



Type: Free radical addition polymerization (chain growth)

Polystyrene — Structural Representation

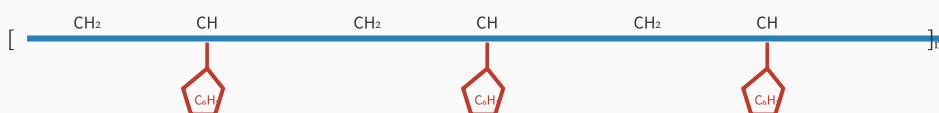


Fig. 5.1 — Polystyrene chain showing alternating CH_2 and $\text{CH(C}_6\text{H}_5\text{)}$ units

5.3 Properties

- Transparent, rigid, colourless polymer.
- Good electrical insulator.
- Brittle (low impact strength).
- Resistant to acids and bases but soluble in benzene and ether.
- Softening point: $\sim 80\text{--}100^\circ\text{C}$.
- Low moisture absorption.

5.4 Applications

Form	Application
Expanded PS (Thermocol)	Packaging, insulation, disposable cups

High-Impact PS (HIPS)	Appliance housings, toys, refrigerator liners
Extruded PS	CD cases, model kits, laboratory ware
PS film	Food packaging, blister packs

5.5 Advantages & Limitations

Advantages	Limitations
Low cost, easy to process	Brittle, low heat resistance
Good optical clarity	Poor UV resistance (yellows)
Excellent electrical insulator	Susceptible to organic solvents
Light weight	Non-biodegradable (environmental concern)

Polyvinyl Chloride (PVC)

6.1 Introduction

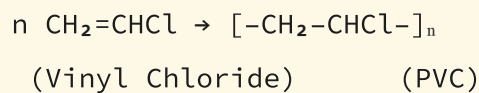
PVC is a thermoplastic polymer produced by free radical polymerization of vinyl chloride monomer (VCM). It is the third most-produced synthetic plastic after polyethylene and polypropylene.

6.2 Preparation

Monomer: Vinyl Chloride ($\text{CH}_2=\text{CHCl}$)

Initiator: Organic peroxides (free radical)

Condition: $50\text{--}70^\circ\text{C}$, suspension or emulsion



Type: Chain growth (free radical addition polymerization)

PVC — Structural Representation and Chlorine Substitution



Fig. 6.1 — PVC chain with Cl atoms attached alternately to the carbon backbone

6.3 Properties

- Hard, rigid polymer in its pure form (unplasticized PVC).
- Good chemical resistance to acids, bases and oils.
- Flame retardant (Cl content makes it self-extinguishing).
- When plasticizers (e.g., DOP) are added → flexible PVC.
- Relatively low cost and easy to process.
- Poor UV resistance; degrades above 60°C releasing HCl.

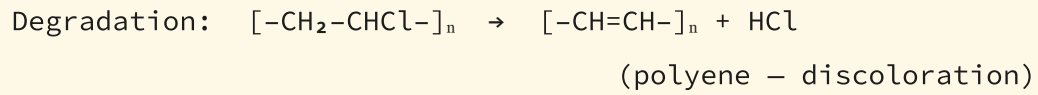
6.4 Forms and Applications

Form	Application
Rigid PVC (uPVC)	Window frames, pipes, fittings, credit cards
Flexible PVC	Wire insulation, garden hoses, raincoats, flooring

PVC foam	Thermal insulation, soundproofing
PVC paste	Coatings, dip moulding, gloves

6.5 Stabilizers Used in PVC Processing

PVC degrades at processing temperatures (~160–200°C) releasing HCl. Stabilizers (lead compounds, organotin, calcium-zinc soaps) are added to prevent degradation.



7 Nylon 6,6 (Polyamide)

7.1 Introduction

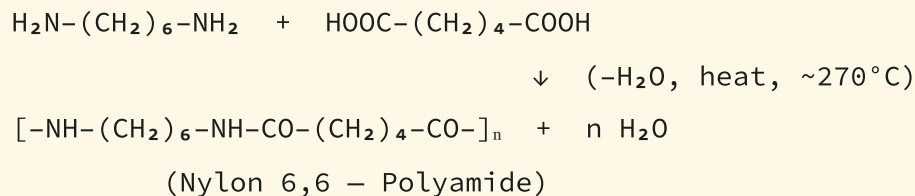
Nylon 6,6 is a thermoplastic polyamide synthesized by step-growth (condensation) polymerization. "6,6" denotes that each monomer contains 6 carbon atoms: hexamethylene diamine (6 carbons) and adipic acid (6 carbons).

7.2 Preparation — Reaction

Monomer A: Hexamethylene diamine $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$

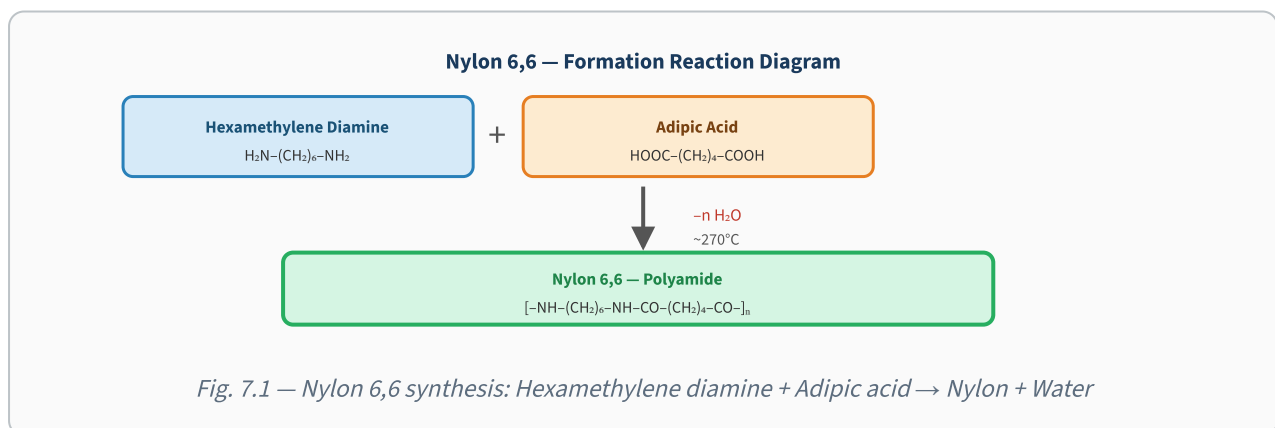
Monomer B: Adipic acid $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$

Reaction:



Linkage formed: Amide bond ($-\text{CO}-\text{NH}-$)

Type: Step-growth condensation polymerization



7.3 Properties

- High tensile strength and toughness.
- Good resistance to wear, abrasion, and fatigue.
- High melting point ($\sim 265^\circ\text{C}$).
- Absorbs moisture (hygroscopic) which affects properties.
- Resistant to oils, fuels, alkalis; attacked by strong acids.
- Good electrical insulator.

7.4 Applications

Sector	Application
Textile / Fibres	Stockings, ropes, parachutes, toothbrush bristles
Engineering	Gears, bearings, bushings, rollers
Electrical	Cable ties, connector housings, switch components
Automotive	Engine covers, air intake manifolds, fuel lines

8.1 Introduction

Bakelite is the first commercially produced synthetic thermosetting polymer, invented by Leo Baekeland in 1907. It is produced by the condensation reaction of phenol and formaldehyde.

8.2 Preparation — Two Stages

Stage 1 — Formation of Novolac (Resole) — Linear Intermediate

Acidic Condition (excess phenol):

Phenol + HCHO → Novolac (linear, thermoplastic)

Alkaline Condition (excess formaldehyde):

Phenol + HCHO → Resole (branched, thermoplastic)

Stage 2 — Curing to Bakelite (Cross-linked Network)

Novolac/Resole + heat (150–200°C) → Bakelite (3D cross-linked network)

Type: Step-growth condensation polymerization

By-product: H₂O

Phenol functionality: $f = 3$ (ortho/para positions active)

Formaldehyde functionality: $f = 2$

Bakelite — Cross-linked Phenol-Formaldehyde Network

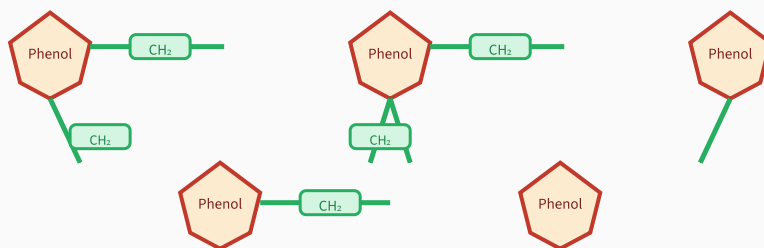


Fig. 8.1 — Bakelite: Phenol rings connected by $\text{-CH}_2\text{-}$ bridges forming rigid 3D network

8.3 Properties

- Hard, rigid, brittle thermosetting polymer.
- Excellent electrical insulator and heat resistant (up to 200°C).
- Resistant to most solvents, weak acids and alkalis.
- Good dimensional stability.

- Dark-coloured (brown/black); cannot be made transparent.

8.4 Applications

- Electrical switches, sockets, plugs, and circuit boards.
- Handles for pots, pans, and utensils.
- Automobile distributor caps and brake linings.
- Billiard balls, cameras, and early telephones.

Elastomers (rubbers) are polymers that exhibit large reversible elastic deformation under stress. They have lightly cross-linked coiled chains that can be stretched to several times their original length and return to the original shape on release of stress. The process of introducing cross-links is called **vulcanization**.

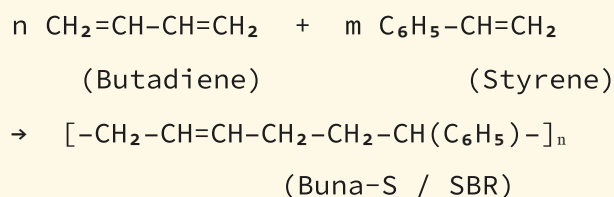
9.1 Buna-S (Styrene-Butadiene Rubber — SBR)

Preparation:

Monomers: Butadiene (75%) + Styrene (25%)

Initiator: Free radical (emulsion polymerization)

Condition: 5°C (cold SBR) or 50°C (hot SBR)



Properties:

- Good abrasion and wear resistance.
- Better aging resistance than natural rubber.
- Good heat resistance (not as good as Neoprene).
- Poor resistance to oils and fuels.
- Good adhesion to steel and fabric.

Applications:

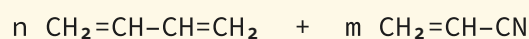
Automobile tyres (largest use — ~70% of SBR production), conveyor belts, shoe soles, gaskets, hoses, and wire insulation.

9.2 Buna-N (Nitrile Butadiene Rubber — NBR)

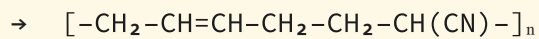
Preparation:

Monomers: Butadiene + Acrylonitrile

Condition: Emulsion polymerization



(Butadiene) (Acrylonitrile)



(Buna-N / NBR)

Properties:

- **Excellent** oil, fuel, and grease resistance (due to $-\text{CN}$ groups).
- Good heat resistance (up to 120°C).
- Poor low-temperature flexibility.
- Good abrasion resistance.

Applications:

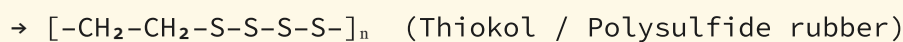
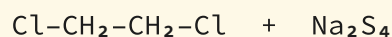
Oil-resistant seals, O-rings, gaskets, fuel hoses, oil well equipment, gloves for handling chemicals.

9.3 Thiokol (Polysulfide Rubber)

Preparation:

Monomers: Ethylene dichloride + Sodium polysulfide

Condition: Aqueous reaction, $\sim 50^\circ\text{C}$



By-product: NaCl

Properties:

- **Excellent** resistance to oils, fuels, solvents, and ozone.
- Good low-temperature flexibility (down to -55°C).
- Strong disagreeable odour (sulfur).
- Poor mechanical properties (lower tensile strength).

Applications:

Fuel tank linings, aircraft fuel hose, sealants for aerospace, chemical-resistant gaskets, printing rolls.

Elastomer Structure — Coiled Chains with Cross-links (Vulcanization)

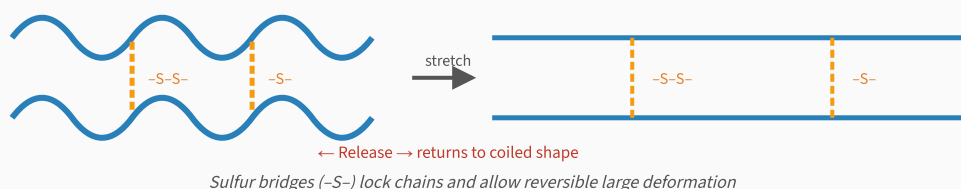


Fig. 9.1 — Elastomer coiled chain structure with sulfur cross-links (vulcanization)

9.4 Elastomers Comparison Table

Property	Buna-S (SBR)	Buna-N (NBR)	Thiokol
Monomers	Butadiene + Styrene	Butadiene + Acrylonitrile	Eth. dichloride + Na ₂ S ₄
Oil Resistance	Poor	Excellent	Excellent
Abrasion Resistance	Excellent	Good	Poor
Heat Resistance	Good	Good (120°C)	Moderate
Low Temp. Flex.	Good	Poor	Excellent (−55°C)
Main Use	Tyres, belts	O-rings, hoses	Aerospace seals

10 Introduction to Fuels & Calorific Value

10.1 Definition of Fuel

A **fuel** is a substance which on combustion (burning in oxygen or air) produces a large amount of heat energy that can be used for practical purposes such as cooking, heating, industrial processes, and power generation.

10.2 Classification of Fuels

Basis	Type	Examples
Physical State	Solid Fuels	Coal, Coke, Wood, Charcoal, Peat
	Liquid Fuels	Petrol, Diesel, Kerosene, Alcohol, Fuel oil
	Gaseous Fuels	Natural gas, LPG, Biogas, Hydrogen, CNG
Origin	Natural Fuels	Coal, Petroleum, Natural gas, Wood
	Manufactured Fuels	Coke, LPG, Charcoal, Biogas
Calorific Value	High CV Fuels	Hydrogen (34,000 kcal/m ³), Natural gas
	Low CV Fuels	Wood, Bagasse, Producer gas

10.3 Calorific Value — Definition and Types

The **Calorific Value (CV)** of a fuel is the amount of heat energy liberated per unit mass (or volume) of a fuel when it is completely burnt in excess of oxygen or air and the products of combustion are cooled to the initial temperature (25°C).

Gross (Higher) Calorific Value (GCV / HCV):

Total heat produced including the heat released by condensation of water vapour formed during combustion. Water is in **liquid** form.

Net (Lower) Calorific Value (NCV / LCV):

Heat produced after deducting the latent heat of steam (water vapour not condensed — remains as steam).

$$\text{NCV} = \text{GCV} - (9H/100) \times \text{Latent heat of steam}$$

$$\text{NCV} = \text{GCV} - (9H \times 587) \text{ kJ/kg}$$

Where: H = % hydrogen in fuel

587 kJ/kg = latent heat of vaporization of water

Or: $\text{NCV} = \text{GCV} - 2454 \times (9H/100) \text{ kJ/kg}$ [if GCV in kJ/kg]

10.4 Bomb Calorimeter — Measurement of GCV

Bomb Calorimeter — Schematic Diagram

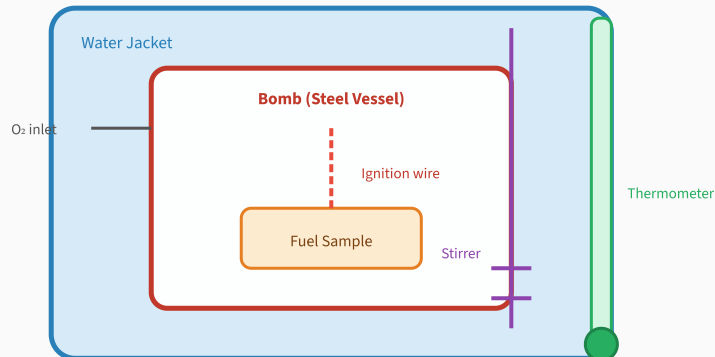


Fig. 10.1 — Bomb Calorimeter for GCV measurement

10.5 Numerical Problems — Calorific Value

Numerical Problem 1: Calculate GCV of Coal

Given: A sample of coal contains C = 80%, H = 6%, S = 2%, O = 5%, N = 1%, Ash = 6%. Calculate GCV by Dulong's formula.

Dulong's Formula:

$$\text{GCV} = (1/100) \times [8080 \times C + 34500 \times (H - O/8) + 2240 \times S] \text{ kcal/kg}$$

Where: C, H, O, S are percentage by weight

- ▶ Step 1: Identify values — C=80, H=6, O=5, S=2
- ▶ Step 2: Calculate $(H - O/8) = 6 - 5/8 = 6 - 0.625 = 5.375$
- ▶ Step 3: $\text{GCV} = (1/100) \times [8080 \times 80 + 34500 \times 5.375 + 2240 \times 2]$
- ▶ Step 4: $= (1/100) \times [646400 + 185437.5 + 4480]$
- ▶ Step 5: $= (1/100) \times 836317.5$
- ▶ **GCV = 8363.18 kcal/kg**

Numerical Problem 2: Calculate NCV from GCV

Given: GCV of coal = 8363 kcal/kg, H = 6%. Find NCV. (Latent heat of steam = 587 kcal/kg)

$$\text{NCV} = \text{GCV} - 9 \times \text{H}/100 \times 587$$

$$\text{NCV} = 8363 - 9 \times (6/100) \times 587$$

$$\text{NCV} = 8363 - 9 \times 0.06 \times 587$$

$$\text{NCV} = 8363 - 317.0$$

$$\text{NCV} = 8046 \text{ kcal/kg}$$

► **NCV = 8046 kcal/kg**

★ **EXAM TIP:** Dulong's formula must be memorized with correct coefficients: 8080 for C, 34500 for H, 2240 for S. NCV formula: $\text{GCV} - 9\text{H} \times 587$.

11 Analysis of Coal

11.1 Proximate Analysis

Proximate Analysis determines the moisture content, volatile matter, fixed carbon, and ash content of coal. It is a physical analysis used for quick assessment of coal quality in industrial settings.

PROCEDURE — STEP BY STEP

1. Moisture Content (M):

1 g coal sample → Dry at 105–110°C for 1 hour in an oven

% Moisture = (Loss in mass / Initial mass) × 100

% M = $[(W_1 - W_2) / W_1] \times 100$

Where: W_1 = initial mass, W_2 = mass after drying

2. Volatile Matter (VM):

Dried coal → Heat at 950°C for 7 min in covered crucible (no air)

% Volatile Matter = (Loss in mass from dried sample / Initial mass) × 100

% VM = $[(W_2 - W_3) / W_1] \times 100$

Where: W_3 = mass after volatile matter removal

3. Ash Content (A):

Coal residue → Burn at 700°C in open crucible until constant mass

% Ash = (Mass of residue / Initial mass) × 100

% A = $(W_4 / W_1) \times 100$

4. Fixed Carbon (FC):

% Fixed Carbon = 100 - (% Moisture + % VM + % Ash)

FC = 100 - M - VM - A

Proximate Analysis — Sequential Steps

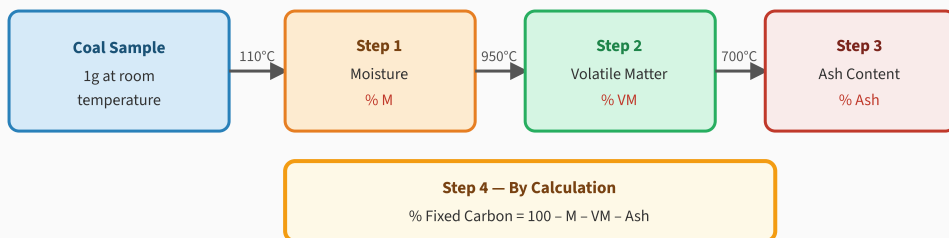


Fig. 11.1 — Sequential procedure for Proximate Analysis of Coal

11.2 Ultimate Analysis

Ultimate Analysis (Elemental Analysis) determines the percentages of Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), Sulfur (S), and Ash in coal. It provides precise chemical composition for calculating calorific value using Dulong's formula.

METHOD — LIEBIG'S COMBUSTION METHOD

Carbon and Hydrogen:

Coal + excess $O_2 \rightarrow CO_2 + H_2O$ (combustion)

CO_2 absorbed in KOH solution $\rightarrow$ mass gives % C

H_2O absorbed in anhydrous $CaCl_2 \rightarrow$ mass gives % H

$$\% C = (12/44) \times (\text{mass of } CO_2 / \text{mass of coal}) \times 100$$

$$\% H = (2/18) \times (\text{mass of } H_2O / \text{mass of coal}) \times 100$$

Nitrogen — Kjeldahl Method:

Coal heated with $H_2SO_4 \rightarrow (NH_4)_2SO_4 \rightarrow NH_3$ (distilled)

NH_3 absorbed in std. HCl $\rightarrow$ back titrated with NaOH

$$\% N = (1.4 \times V \times N) / \text{mass of coal}$$

Where: V = vol. of acid, N = normality

Sulfur:

Combustion products $\rightarrow SO_3$ absorbed $\rightarrow BaSO_4$ precipitate

$$\% S = (32/233) \times (\text{mass of } BaSO_4 / \text{mass of coal}) \times 100$$

Oxygen:

$$\% O = 100 - (\% C + \% H + \% N + \% S + \% \text{Ash})$$

11.3 Comparison — Proximate vs Ultimate Analysis

Feature	Proximate Analysis	Ultimate Analysis
Determines	M, VM, FC, Ash	C, H, O, N, S, Ash
Type	Physical	Chemical (elemental)
Accuracy	Quick/rough estimate	Precise
Use	Industrial assessment	CV calculation, research
Method	Heating in oven/furnace	Combustion + chemical analysis
CV formula use	Not directly	Yes (Dulong's formula)

12 Petroleum Refining, Octane & Cetane Numbers

12.1 Refining of Petroleum — Fractional Distillation

*Petroleum (crude oil) is a naturally occurring complex mixture of hydrocarbons. **Refining** involves separating these fractions based on differences in their boiling points by fractional distillation in a tall fractionating column.*

Petroleum Refinery — Fractional Distillation Column

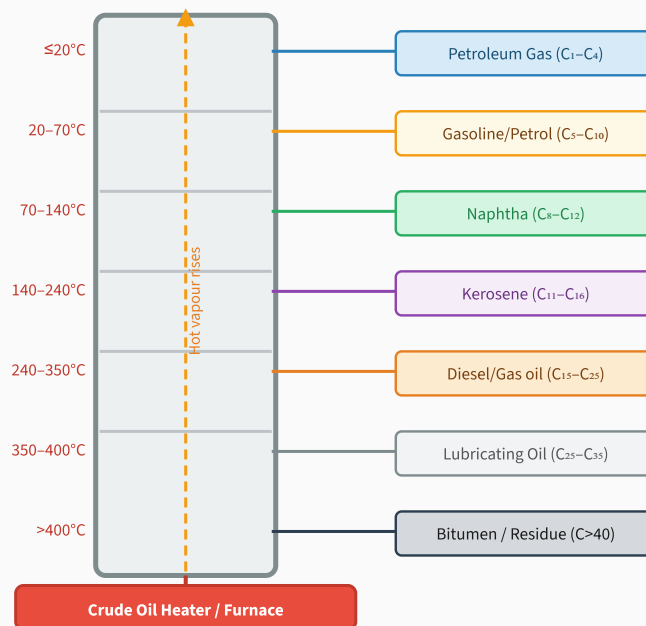


Fig. 12.1 — Petroleum fractional distillation column with temperature zones and fractions

12.2 Octane Number

*The **Octane Number (ON)** is a measure of the antiknock property of a petrol/gasoline. It is defined as the percentage (by volume) of iso-octane in a mixture of iso-octane and n-heptane that has the same knocking tendency as the given fuel under standard test conditions.*

- **Iso-octane (2,2,4-trimethylpentane):** Assigned ON = 100 (excellent antiknock property).
- **n-Heptane:** Assigned ON = 0 (very poor antiknock property).
- Higher octane number → better antiknock → suitable for high-compression engines.
- **Knocking** occurs when fuel ignites before the spark plug fires, causing engine damage.

Octane Number = % of iso-octane in (iso-octane + n-heptane) mixture
that matches fuel's knock resistance

Example: $ON = 87 \rightarrow$ fuel equivalent to 87% iso-octane + 13% n-heptane

Methods to improve Octane Number:

- Adding antiknock agents (formerly TEL – tetraethyl lead; now MTBE, ethanol).
- Reforming (cyclization and aromatization of naphtha).
- Catalytic cracking (converts long-chain hydrocarbons to branched short-chain).
- Isomerization of straight-chain to branched hydrocarbons.

12.3 Cetane Number

The **Cetane Number (CN)** is a measure of the ignition quality (ignition delay) of diesel fuel. It is the percentage by volume of cetane (hexadecane, $C_{16}H_{34}$) in a mixture of cetane and alpha-methylnaphthalene that has the same ignition delay as the test fuel.

- **Cetane (n-hexadecane):** CN = 100 (ignites readily — short delay).
- **Alpha-methylnaphthalene:** CN = 0 (long ignition delay).
- Higher cetane number $\rightarrow$ shorter ignition delay $\rightarrow$ smoother combustion $\rightarrow$ less noise.
- Diesel engines require $CN \geq 45$; good diesel fuel: CN = 50–55.

12.4 Comparison — Octane vs Cetane Number

Property	Octane Number	Cetane Number
Applicable to	Petrol / Gasoline (SI engines)	Diesel (CI engines)
Measures	Antiknock property	Ignition quality (delay)
Standard (100)	Iso-octane	n-Hexadecane (Cetane)
Standard (0)	n-Heptane	Alpha-methylnaphthalene
Higher value means	Better antiknock (resist premature ignition)	Shorter ignition delay (faster ignition)
Ideal value	>90 for premium petrol	45–55 for good diesel
Relation	$ON + CN \approx 100$ (approximate inverse relationship)	

★ **EXAM TIP:** Octane number is for petrol (SI engine); Cetane number is for diesel (CI engine). This distinction is frequently asked. Higher octane = better petrol; higher cetane = better diesel.

13 Alternative Fuels

Alternative fuels are fuels derived from non-petroleum sources and used to replace conventional fossil fuels. They are generally cleaner, renewable, or more sustainable than traditional fuels.

13.1 Propane (LPG Component)

Chemical formula: C_3H_8 | **Calorific value:** ~50,000 kJ/kg

Combustion: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O + \text{Heat}$

Source: Natural gas processing and petroleum refining.

- Clean burning — low particulate emissions.
- Stored as liquid under pressure (LPG cylinders).
- Used in domestic cooking, heating, and autogas.
- Higher energy density than CNG but lower than diesel.

13.2 Methanol (Methyl Alcohol)

Chemical formula: CH_3OH | **CV:** ~22,700 kJ/kg

Production: $CO + 2H_2 \rightarrow CH_3OH$ (from synthesis gas, coal)

Combustion: $2CH_3OH + 3O_2 \rightarrow 2CO_2 + 4H_2O + \text{Heat}$

- High octane number (ON ~ 107) — excellent antiknock property.
- Burns more completely than petrol → lower CO emissions.
- Toxic (poisonous); corrosive to some metals and plastics.
- Lower energy density than petrol (half the CV).
- Used as racing fuel, fuel cell feedstock, and as M85 blend (85% methanol + 15% petrol).

13.3 Ethanol (Ethyl Alcohol)

Chemical formula: C_2H_5OH | **CV:** ~29,700 kJ/kg

Production (Fermentation):

$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (yeast catalysis)
(Glucose) (Ethanol)

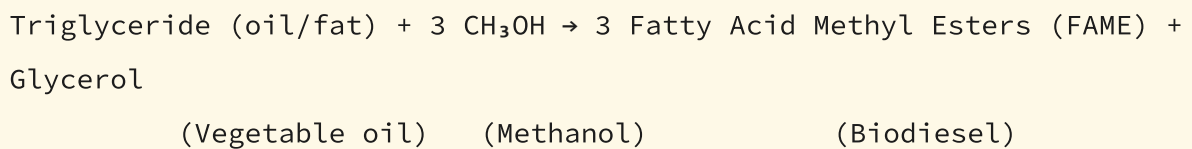
Combustion: $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O} + \text{Heat}$

- Renewable (from sugarcane, corn, cellulose).
- High octane number (ON ~ 110) — good antiknock fuel.
- Contains oxygen (35%) → more complete combustion → fewer hydrocarbons in exhaust.
- Used as E10 (10% ethanol in petrol) and E85 blends.
- Hygroscopic (absorbs water) — causes phase separation in blends.
- India's Ethanol Blending Programme (EBP) — 20% blending target by 2025.

13.4 Biofuels — Biodiesel

Biodiesel is a renewable fuel made from vegetable oils, animal fats, or recycled cooking oils by a process called **transesterification**. It can be used directly or blended with petroleum diesel (B5, B20, B100).

Transesterification:



Catalyst: NaOH or KOH (alkaline)

Conditions: 60–70°C, ~1 hour

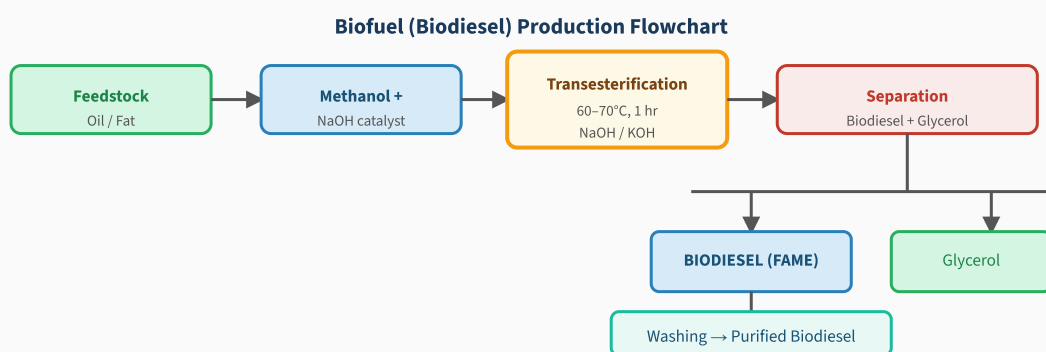


Fig. 13.1 — Biodiesel production by transesterification of vegetable oils

13.5 Comparison — Conventional vs Alternative Fuels

Property	Petrol/Diesel	Methanol	Ethanol	Biodiesel
Source	Petroleum (non-renewable)	Synthesis gas / Coal	Biomass (renewable)	Vegetable oils (renewable)
CV (kJ/kg)	~44,000	~22,700	~29,700	~37,000

Octane No.	87–95	~107	~110	50–60 (CN)
CO ₂ balance	Positive (adds CO ₂)	Partial	Near-zero	Near-zero
Toxicity	Moderate	High (toxic)	Low	Low
Blending	—	M85	E10, E85	B5, B20, B100

14 Questions & Answers

Part A — 2-Mark Questions (Short Answers)

Q1. Define a polymer and give two examples.

2 Marks

A **polymer** is a large macromolecule composed of many repeating structural units called monomers, joined together by covalent bonds through polymerization. The prefix "poly" means many and "mer" means unit. Polymers have very high molecular weights ranging from thousands to millions. They can be natural (starch, cellulose, natural rubber) or synthetic (Nylon, PVC, Polystyrene). Examples: (1) Polyethylene — formed from ethylene monomers; (2) Nylon 6,6 — formed from hexamethylene diamine and adipic acid.

Q2. What is functionality of a monomer? What is the functionality of phenol?

2 Marks

The **functionality** of a monomer is the number of reactive sites or functional groups available per monomer molecule for forming bonds during polymerization. It determines the nature of the polymer formed — linear ($f=2$), branched ($f=3$), or cross-linked ($f>3$). Phenol has three reactive positions — one ortho, another ortho, and one para — all available for reaction with formaldehyde. Therefore, the functionality of phenol is **3**. This trifunctionality enables phenol to form a three-dimensional cross-linked network in Bakelite.

Q3. Define Calorific Value. Distinguish between GCV and NCV.

2 Marks

Calorific Value is the amount of heat energy released per unit mass (or volume) of a fuel when it is completely burned in excess oxygen and the products are cooled to the initial temperature. **Gross Calorific Value (GCV)** includes the latent heat released when water vapour condenses to liquid water. **Net Calorific Value (NCV)** = GCV minus the latent heat of vaporization of the water produced. NCV represents the practically usable heat since steam does not always condense in real applications. Formula: $NCV = GCV - 9H \times 587 \text{ kcal/kg}$.

Q4. Define Octane Number and Cetane Number.

2 Marks

Octane Number (ON): It is the percentage by volume of iso-octane (2,2,4-trimethylpentane) in a mixture with n-heptane that has the same antiknock property as the given fuel. Iso-octane = ON 100; n-heptane = ON 0. Used for petrol/gasoline. Higher ON = better antiknock = suitable for high-compression engines. **Cetane Number (CN):** It is the percentage by volume of cetane (n-hexadecane) in a mixture with alpha-methylnaphthalene having the same ignition delay as the test diesel fuel. Cetane = CN 100; alpha-methylnaphthalene = CN 0. Higher CN = shorter ignition delay = smoother diesel combustion. Good diesel fuel CN = 45–55.

Q5. What is vulcanization? Why is it done?**2 Marks**

Vulcanization is the process of introducing cross-links between polymer chains of natural or synthetic rubber by heating with sulfur (or sulfur compounds). The process was discovered by Charles Goodyear (1839). Without vulcanization, rubber is sticky, weak, and deforms permanently. After vulcanization: (1) Cross-links (–S–S–) restrict chain slippage, (2) Elasticity improves — rubber can be stretched repeatedly without permanent deformation, (3) Tensile strength and hardness increase, (4) Resistance to heat, abrasion, and solvents improves. The degree of cross-linking determines the hardness: lightly vulcanized = elastic; heavily vulcanized = hard rubber (ebonite).

Q6. What is proximate analysis of coal? Name the parameters determined.**2 Marks**

Proximate Analysis of coal is a physical analysis method that determines the approximate composition of coal in terms of its major components. It is simpler and faster than ultimate analysis and is widely used in industrial settings for quick quality assessment. The four parameters determined are: (1) **Moisture Content (M)**: by heating at 105–110°C; (2) **Volatile Matter (VM)**: by heating at 950°C in absence of air; (3) **Ash Content (A)**: by complete combustion at 700°C; (4) **Fixed Carbon (FC)**: calculated as $FC = 100 - M - VM - \text{Ash}$. Higher FC and lower ash indicate better quality coal (e.g., anthracite).

Part B — 5-Mark Questions (Long Answers)

Q7. Explain the mechanism of chain growth polymerization with examples.

5 Marks

Chain Growth (Addition) Polymerization involves the sequential and rapid addition of monomer units possessing carbon-carbon double bonds ($C=C$) to a growing active chain. No by-product is formed. The molecular weight increases rapidly.

Mechanism — Three Steps:

1. Initiation: An initiator (e.g., benzoyl peroxide) undergoes homolytic cleavage on heating to form free radicals ($R\cdot$). This radical attacks the double bond of a monomer (e.g., styrene), creating an active monomer radical. The π -bond breaks and one electron pairs with the radical while the other becomes a new radical centre on the terminal carbon.

2. Propagation: The monomer radical attacks another monomer molecule, extending the chain. This process repeats rapidly (milliseconds) — each step adds one monomer unit. The chain grows in a head-to-tail fashion. The active centre ($\cdot$) is always at the chain end.

3. Termination: The active chain ends when: (a) *Combination* — two active chain radicals couple together to form one dead polymer; (b) *Disproportionation* — hydrogen transfer between two chains produces two dead polymers, one with a terminal double bond.

Reactions: $R\cdot + CH_2=CHX \rightarrow R-CH_2-CHX\cdot$ (initiation); $R-(CH_2CHX)_n\cdot + CH_2=CHX \rightarrow R-(CH_2CHX)_{n+1}\cdot$ (propagation); $2 R-(CH_2CHX)_n\cdot \rightarrow R-(CH_2CHX)_{2n}-R$ (termination by combination).

Examples: Polystyrene (from styrene), PVC (from vinyl chloride), Polyethylene (from ethylene), Poly(methyl methacrylate) — PMMA (from MMA). These are all addition polymers produced via free radical chain growth mechanism. Chain growth is much faster than step growth, producing high molecular weight polymer within seconds.

Q8. Differentiate chain growth and step growth polymerization. Give two examples of each.

5 Marks

Chain Growth vs Step Growth Polymerization:

1. Monomer type: Chain growth uses monomers with $C=C$ double bonds (unsaturated). Step growth uses bifunctional monomers containing functional groups like $-OH$, $-COOH$, $-NH_2$, $-COCl$ that react with each other.

2. By-product: Chain growth — no by-product formed (addition reaction). Step growth — a small molecule by-product is eliminated (H_2O , HCl , CH_3OH) in every step.

3. Rate of polymerization: Chain growth is very fast (milliseconds to seconds for high MW). Step growth is slow (hours required to achieve high molecular weight).

4. Molecular weight development: In chain growth, high MW polymer forms right from the beginning; monomer is consumed gradually. In step growth, MW increases slowly; intermediate oligomers (dimers, trimers) accumulate initially; very high conversion (>99%) needed for high MW.

5. Initiator: Chain growth requires initiator (peroxide, UV, ionic catalyst). Step growth does not need initiator; heat and catalyst (acid/base) sufficient.

6. Mechanism: Chain growth — Initiation, Propagation, Termination (IPT). Step growth — stepwise condensation between any two functional groups.

7. Molecular weight distribution: Chain growth gives broad MW distribution. Step growth gives narrower distribution (Schulz-Flory distribution).

Examples of Chain Growth: Polystyrene (styrene), PVC (vinyl chloride). **Examples of Step Growth:** Nylon 6,6 (hexamethylene diamine + adipic acid), Bakelite (phenol + formaldehyde).

Q9. Describe the preparation, properties, and applications of Nylon 6,6.

5 Marks

Nylon 6,6 is a synthetic thermoplastic polyamide polymer produced by condensation polymerization. The name "6,6" indicates that both the diamine (hexamethylene diamine) and the diacid (adipic acid) each contain six carbon atoms.

Preparation: Nylon 6,6 is prepared by the reaction of hexamethylene diamine $[H_2N-(CH_2)_6-NH_2]$ with adipic acid $[HOOC-(CH_2)_4-COOH]$. The reaction occurs at $\sim 270^\circ C$ under inert atmosphere with removal of water. The two monomers first form a salt called "Nylon salt" which on further heating undergoes condensation to give Nylon 6,6 polymer. The linkage formed is an amide bond $(-CO-NH-)$.

Reaction: $n H_2N-(CH_2)_6-NH_2 + n HOOC-(CH_2)_4-COOH \rightarrow [-NH-(CH_2)_6-NH-CO-(CH_2)_4-CO-]_n + n H_2O$

Properties: (1) High tensile strength and good elastic recovery. (2) Excellent abrasion and wear resistance. (3) High melting point ($\sim 265^\circ C$). (4) Good chemical resistance to oils, fuels, alkalis, and most organic solvents. (5) Hygroscopic — absorbs moisture from air which can reduce mechanical properties. (6) Good electrical insulator. (7) Can be melt-spun into fibres or injection-moulded into parts.

Applications: Textile fibres (stockings, lingerie, parachutes, fishing lines, ropes), engineering parts (gears, bearings, bushings, cams, pulleys), electrical components (connector housings, circuit breakers, cable ties), automotive (engine covers, air intake manifolds, fuel lines, radiator fans), and consumer goods (toothbrush bristles, zippers).

Q10. Explain proximate and ultimate analysis of coal with their significance.

5 Marks

Proximate Analysis: This is a simple, physical analysis of coal that determines four key components:

(1) **Moisture Content:** 1g coal is heated at $105-110^\circ C$ for 1 hour. Loss in weight = moisture. High moisture reduces calorific value. $\%M = (W_1 - W_2)/W_1 \times 100$.

(2) **Volatile Matter:** The dried coal sample is heated at $950^\circ C$ for 7 minutes in a covered crucible (no air). Loss in weight = volatile matter. High VM $\rightarrow$ easy ignition but smoky flame. $\%VM = (W_2 - W_3)/W_1 \times 100$.

(3) **Ash Content:** The remaining coal is burned completely at 700°C in open crucible. Residue = ash. High ash reduces CV and causes clinker problems. $\%A = W_4/W_1 \times 100$.

(4) **Fixed Carbon:** Calculated as $FC = 100 - M - VM - Ash$. High FC indicates high quality coal and high calorific value.

Ultimate Analysis: This is a precise elemental (chemical) analysis determining the percentages of C, H, O, N, S, and Ash in coal.

(1) **Carbon and Hydrogen:** Coal is burned; CO₂ absorbed in KOH gives %C; H₂O absorbed in CaCl₂ gives %H.

(2) **Nitrogen:** Kjeldahl method — coal digested with H₂SO₄, NH₃ distilled and titrated.

(3) **Sulfur:** Combustion products oxidized; BaSO₄ precipitate weighed. High S is undesirable (causes acid rain).

(4) **Oxygen:** Calculated by difference: $\%O = 100 - C - H - N - S - Ash$.

Significance: Proximate analysis → rapid industrial quality assessment. Ultimate analysis → precise calorific value calculation (Dulong's formula), power plant efficiency assessment, and environmental compliance.

Q11. Explain petroleum refining and the significance of Octane and Cetane numbers.**5 Marks**

Petroleum Refining: Crude petroleum is a complex mixture of hydrocarbons of varying chain lengths (C_1 to C_{50+}). Refining separates these into useful fractions by **fractional distillation** in a tall column (30–50 m high) with trays at different heights. Crude oil is heated in a furnace to $\sim 400^\circ\text{C}$ and the vapours enter the column.

Lighter fractions with lower boiling points rise to the top while heavier fractions condense lower down. The main fractions obtained (bottom to top): (1) Residue/Bitumen ($>400^\circ\text{C}$) — road surfacing, waterproofing. (2) Lubricating oil ($350\text{--}400^\circ\text{C}$) — engine lubrication. (3) Gas oil/Diesel ($240\text{--}350^\circ\text{C}$) — diesel engines, power plants. (4) Kerosene ($140\text{--}240^\circ\text{C}$) — jet fuel, lamps, stoves. (5) Naphtha ($70\text{--}140^\circ\text{C}$) — petrochemical feedstock. (6) Petrol/Gasoline ($20\text{--}70^\circ\text{C}$) — automobiles. (7) Petroleum gas ($<20^\circ\text{C}$) — LPG cylinders.

Octane Number: Petrol in high-compression SI (spark ignition) engines can undergo "knocking" — premature self-ignition before the spark fires, causing engine damage and loss of power. The Octane Number (ON) measures resistance to knocking. ON is the % of iso-octane (ON=100) in a mixture with n-heptane (ON=0) matching the fuel's knock resistance. Higher ON (e.g., 95 for premium petrol) → better antiknock → suitable for high-performance engines. Methods to improve ON: adding MTBE, reforming, isomerization, and catalytic cracking.

Cetane Number: Diesel in CI (compression ignition) engines must ignite spontaneously when injected (no spark plug). CN measures ignition delay — shorter delay = better combustion. CN = % of cetane (n-hexadecane, CN=100) in a mixture with alpha-methylnaphthalene (CN=0) with same ignition delay. Good diesel fuel has CN = 45–55. Low CN causes rough start, noise ("diesel knock"), and incomplete combustion. High CN gives smooth, quiet engine operation.

Q12. Write notes on Buna-S, Buna-N, and Thiokol rubber with preparation and applications.**5 Marks**

Buna-S (SBR — Styrene-Butadiene Rubber): Prepared by free radical emulsion co-polymerization of butadiene (75%) and styrene (25%) at $5\text{--}50^\circ\text{C}$. The name "Buna" comes from butadiene + Na (sodium, the catalyst used historically) and S for styrene. It is the most widely produced synthetic rubber. Properties: Good abrasion resistance, better aging resistance than NR, good heat and flex fatigue resistance. Applications: Automobile tyres (largest use), conveyor belts, shoe soles, gaskets, flooring, wire insulation.

Buna-N (NBR — Nitrile Butadiene Rubber): Prepared by co-polymerization of butadiene (60–80%) and acrylonitrile (20–40%) by emulsion polymerization. The CN (nitrile) groups give it excellent oil resistance. Properties: Excellent oil, fuel, and grease resistance (best among elastomers), good heat resistance up to 120°C , good abrasion resistance, poor low-temperature flexibility. Applications: Oil-resistant O-rings, seals, gaskets, fuel hoses, fuel tanks, gloves for chemical handling, oil well equipment.

Thiokol (Polysulfide Rubber): Prepared by reaction of ethylene dichloride and sodium polysulfide (Na_2S_x) in aqueous suspension. By-product: NaCl . The polymer chain contains $-\text{S}-\text{S}-$ linkages. Properties: Exceptional resistance to oils, fuels, solvents, and ozone; excellent low-temperature flexibility (down to -55°C); poor mechanical strength; strong unpleasant odour due to sulfur. Applications: Fuel tank linings, aircraft fuel hoses, sealants for aircraft windows and doors (aerospace sealants), printing rolls, and chemical-resistant gaskets.

Numerical Problems with Solutions

Numerical Problem 3: Calorific Value Calculation

Problem: The ultimate analysis of a coal sample gives: C = 75%, H = 5%, O = 8%, N = 2%, S = 3%, Ash = 7%. Calculate GCV and NCV by Dulong's formula. (Latent heat of steam = 587 kcal/kg)

Dulong's Formula:

$$\text{GCV} = (1/100) \times [8080 \times C + 34500 \times (H - O/8) + 2240 \times S]$$

- ▶ Step 1: Given — C=75, H=5, O=8, S=3
- ▶ Step 2: Calculate $(H - O/8) = 5 - 8/8 = 5 - 1 = 4.0$
- ▶ Step 3: $\text{GCV} = (1/100) \times [8080 \times 75 + 34500 \times 4.0 + 2240 \times 3]$
- ▶ Step 4: $= (1/100) \times [606000 + 138000 + 6720]$
- ▶ Step 5: $= (1/100) \times 750720$
- ▶ **GCV = 7507.2 kcal/kg**
- ▶ Step 6: $\text{NCV} = \text{GCV} - 9H \times 587/100 = 7507.2 - 9 \times 5 \times 587/100$
- ▶ Step 7: $= 7507.2 - 264.15 = \text{NCV} = 7243.05 \text{ kcal/kg}$

Numerical Problem 4: Coal Analysis Data

Problem: During proximate analysis of coal, the following data was obtained: Initial mass = 2.0 g; Mass after drying at 110°C = 1.88 g; Mass after heating at 950°C = 1.48 g; Mass of ash residue = 0.28 g. Calculate % Moisture, % VM, % Ash, and % Fixed Carbon.

- ▶ Step 1: % Moisture $= (2.0 - 1.88)/2.0 \times 100 = 0.12/2.0 \times 100 = \mathbf{6\%}$
- ▶ Step 2: % Volatile Matter $= (1.88 - 1.48)/2.0 \times 100 = 0.40/2.0 \times 100 = \mathbf{20\%}$
- ▶ Step 3: % Ash $= 0.28/2.0 \times 100 = \mathbf{14\%}$
- ▶ Step 4: % Fixed Carbon $= 100 - 6 - 20 - 14 = \mathbf{60\%}$
- ▶ **Result:** M=6%, VM=20%, Ash=14%, FC=60%

Key Polymer Definitions

- **Polymer** = many monomers linked by covalent bonds
- **Degree of polymerization (n)** = no. of repeat units
- **Functionality (f)** = reactive sites per monomer
- **Thermoplastic** = softens on heating (reversible)
- **Thermosetting** = permanent on heating (irreversible)
- **Elastomer** = large elastic deformation possible

Polymerization Types

- **Chain growth:** C=C monomers, fast, no by-product
- **Step growth:** functional groups, slow, by-product (H₂O)
- Mechanism: Initiation → Propagation → Termination
- PS, PVC, PE → chain growth
- Nylon, Bakelite, Polyester → step growth

Important Polymer Reactions

Polystyrene: $n \text{ C}_6\text{H}_5\text{CH}=\text{CH}_2 \rightarrow [-\text{CH}_2-\text{CH}(\text{C}_6\text{H}_5)-]_n$

PVC: $n \text{ CH}_2=\text{CHCl} \rightarrow [-\text{CH}_2-\text{CHCl}-]_n$

Nylon 6,6: diamine + diacid → polyamide + H₂O

Bakelite: phenol + HCHO → 3D cross-linked + H₂O

Buna-S: butadiene + styrene → SBR

Fuel Calorific Value Formulas**Dulong's Formula:**

$\text{GCV} = (1/100) \times [8080C + 34500(\text{H}-\text{O}/8) + 2240S]$
kcal/kg

NCV Formula:

$\text{NCV} = \text{GCV} - 9H \times 587/100$

or: $\text{NCV} = \text{GCV} - 2454 \times 9H/100$ (in kJ/kg)

Coal Analysis Key Points

- Proximate: M (110°C), VM (950°C), Ash (700°C), FC (by calc)
- Ultimate: C, H, O, N, S, Ash (elemental)
- $\text{FC} = 100 - \text{M} - \text{VM} - \text{Ash}$
- Higher FC → better coal quality
- Higher Ash → lower CV, more clinker
- Dulong's uses Ultimate analysis data

Octane & Cetane Numbers

- Octane No. → Petrol (SI engine) → antiknock
- Cetane No. → Diesel (CI engine) → ignition delay
- Iso-octane = ON 100; n-heptane = ON 0
- Cetane = CN 100; α-methylnaphthalene = CN 0
- Higher ON = better petrol quality
- Higher CN = shorter ignition delay = better diesel
- Good petrol: ON > 87; Good diesel: CN = 45–55

Elastomers Summary**Alternative Fuels Comparison**

• Buna-S = butadiene + styrene → tyres • Buna-N = butadiene + acrylonitrile → oil-resistant seals • Thiokol = ethylene dichloride + Na₂S₄ → aerospace sealants • Vulcanization: cross-links with S → improved elasticity & strength	• Methanol: ON~107, toxic, CV~22700 kJ/kg • Ethanol: ON~110, renewable, CV~29700 kJ/kg • Biodiesel: from transesterification of oils, B5/B20 • Propane: LPG component, CV~50000 kJ/kg • Transesterification: oil + CH₃OH → FAME + glycerol
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At-a-Glance: Commercial Polymers

Polymer	Monomer(s)	Type	Reaction	Key Property	Use
Polystyrene (PS)	Styrene	Thermoplastic	Chain growth	Transparent, rigid, brittle	Packaging, thermocol
PVC	Vinyl chloride	Thermoplastic	Chain growth	Flame-retardant, versatile	Pipes, wire insulation
Nylon 6,6	HMD + Adipic acid	Thermoplastic	Step growth	High strength, wear-resistant	Fibres, gears
Bakelite	Phenol + Formaldehyde	Thermosetting	Step growth	Rigid, infusible, insulator	Electrical fittings
Buna-S (SBR)	Butadiene + Styrene	Elastomer	Chain growth	Good abrasion resistance	Tyres, belts
Buna-N (NBR)	Butadiene + Acrylonitrile	Elastomer	Chain growth	Oil/fuel resistant	O-rings, fuel hoses
Thiokol	Eth. dichloride + Na ₂ S ₄	Elastomer	Step growth	Solvent/ozone resistant	Aerospace sealants

Petroleum Fractions — Memory Aid

Fraction	Boiling Range	Carbon Range	Uses
Petroleum Gas	<20°C	C ₁ –C ₄	LPG, cooking
Gasoline/Petrol	20–70°C	C ₅ –C ₁₀	Automobile fuel (SI engines)
Naphtha	70–140°C	C ₈ –C ₁₂	Petrochemical feedstock
Kerosene	140–240°C	C ₁₁ –C ₁₆	Jet fuel, lamps
Diesel / Gas oil	240–350°C	C ₁₅ –C ₂₅	CI engines, power plants
Lubricating oil	350–400°C	C ₂₅ –C ₃₅	Engine lubrication
Bitumen/Residue	>400°C	C>40	Road surfacing, roofing

★ **EXAM TIP:** For exam: Always write Dulong's formula coefficients correctly (8080C, 34500H, 2240S).

Memorize the three steps of chain growth (IPT). Know one reaction for each polymer. Practise at least 2 numerical problems on calorific value.

Primary Textbooks

1. **Jain, P.C. and Jain, M.** — *Engineering Chemistry*, 16th Edition, Dhanpat Rai Publishing Company, New Delhi, 2013.
Chapters: Polymer Chemistry (Unit VII), Fuels (Unit IX), Coal Analysis and Petroleum Refining.
2. **Atkins, P., de Paula, J., and Keeler, J.** — *Atkins' Physical Chemistry*, 10th Edition, Oxford University Press, 2010.
Chapters: Macromolecules and Polymers, Thermodynamics of Combustion.

Reference Books

3. **Taylor, H.F.W.** — *Cement Chemistry*, 2nd Edition, Thomas Telford Publishing, London, 1997.
(Reference for material chemistry and industrial chemical processes.)
4. **Shaw, D.J.** — *Introduction to Colloids and Surface Chemistry*, 4th Edition, Butterworth-Heinemann, 1992.
(Relevant for understanding polymer interfaces and emulsion polymerization.)
5. **Billmeyer Jr., Fred W.** — *Textbook of Polymer Science*, 3rd Edition, John Wiley & Sons, Inc., 1984.
A comprehensive reference for polymer structure, morphology, molecular weight distribution, and polymerization mechanisms. Chapters 2–6 are particularly relevant.

Additional Reading & Online Resources

6. **Seymour, R.B. and Carraher, C.E.** — *Polymer Chemistry: An Introduction*, 5th Edition, Marcel Dekker, 2000.
7. **Gowariker, V.R., Viswanathan, N.V., and Sreedhar, J.** — *Polymer Science*, New Age International Publishers, 2005.
(Excellent Indian author reference for B.Tech level.)
8. **Datta, R.L. and Chatterjee, N.N.** — *Engineering Chemistry*, Oriental Longman, Kolkata, 2007.

Important Formulae — Final Summary

1. Molecular Weight of Polymer = $M_0 \times n$
2. Average Functionality: $f_{avg} = \frac{\sum(f_i \times N_i)}{\sum N_i}$
3. Dulong's Formula (GCV):
$$GCV = (1/100) \times [8080C + 34500(H - O/8) + 2240S] \quad \text{kcal/kg}$$
4. NCV (Net Calorific Value):
$$NCV = GCV - 9H \times 587 / 100 \quad \text{kcal/kg}$$

$$\text{NCV} = \text{GCV} - 2454 \times 9\text{H}/100 \quad \text{kJ/kg}$$

5. Carbon in Ultimate Analysis:

$$\%C = (12/44) \times (\text{mass CO}_2 / \text{mass coal}) \times 100$$

6. Hydrogen in Ultimate Analysis:

$$\%H = (2/18) \times (\text{mass H}_2\text{O} / \text{mass coal}) \times 100$$

7. Fixed Carbon (Proximate):

$$\%FC = 100 - \%M - \%VM - \%Ash$$

8. Octane Number: % iso-octane in (iso-octane + n-heptane) mix

9. Cetane Number: % cetane in (cetane + α -methylnaphthalene) mix

This study material has been prepared for B.Tech 1st Year Engineering Chemistry — Unit III: Polymers and Fuel Chemistry.

All reactions, formulas, and diagrams are exam-oriented and aligned with university syllabus requirements.

For comprehensive study, students are advised to refer to the textbooks listed above.

Best Wishes for Your Examinations!